Connections: how to invent crystal maths

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$X/\mathbb{F}_p$ smooth proj. var. , $\tilde{X}/\mathbb{Z}_p$ smooth proj. lift

$\rightsquigarrow H^{\bullet}_{crys}(X/\mathbb{Z}_p) := H^{\bullet}_{dR}(\tilde{X}/\mathbb{Z}_p) = \mathcal{E}_{\tilde{s}}$

Suppose $\exists$ nice moduli space $S/\mathbb{Z}_p$ smooth of "proj. sm. vars" ($\mathfrak{X} \to S$)

$X \leftrightsquigarrow s \in S(\mathbb{F}_p)$

$\tilde{X} \leftrightsquigarrow \tilde{s} \in S(\mathbb{Z}_p)$ lift of s $\quad \left(\mathcal{E} := H^{\bullet}_{dR}(\mathfrak{X}/S) \text{ v.b.}, \nabla_{GM} \right)$

<u>Q</u>. can we canonically identify $\mathcal{E}_{\tilde{s}}$ for various lifts $\tilde{s}$?

$\longrightarrow \mathcal{E}$

$X \xrightarrow{f} S$ map of schemes.

$$\begin{tikzcd} \Delta_X \arrow[r, hook] \arrow[dr, "\text{cl.}"'] & X \times_S X \\ & X^{(2)} = V(I^2_{\Delta_X}) \arrow[u, "\text{loc. closed}"'] \end{tikzcd}$$

square zero thickening

$\exists$ canonical isom. $\Omega_{X/S} \simeq I_{X \hookrightarrow X^{(2)}}$ as $\mathcal{O}_X$-mod.

$$\begin{tikzcd} X \arrow[r] \arrow[dr] & X \times_S X \times_S X \\ & X^{(3)} = V(I^2_{X \hookrightarrow X \times_S X \times_S X}) \arrow[u] \end{tikzcd}$$

<u>Th</u>. Given $\mathcal{F}/X$ quasi-coherent, TFAE:

① a connection (satisfying Leibniz)

$$\nabla : \mathcal{F} \to \mathcal{F} \otimes \Omega_{X/S}$$

② a descent datum for $\mathcal{F}$ on

$$X \underset{p_2}{\overset{p_1}{\leftleftarrows}} X^{(2)} \overset{p_{12}, p_{13}, p_{23}}{\leftarrow} X^{(3)}$$

i.e. an isom. $p_1^* \mathcal{F} \simeq p_2^* \mathcal{F}$ satisfying cocycle condition on $X^{(3)}$.

②' isom. $p_1^* \mathcal{F} \simeq p_2^* \mathcal{F}$ that is the identity $\mathrm{id}_{\mathcal{F}}$ when restr. to $X \hookrightarrow X^{(2)}$.

Define full subcat. $\mathrm{Crys}^2(X/S)$ of diagrams

$$\begin{tikzcd} U \arrow[r, hook, "\text{closed}"', "\text{sq. zero}"] \arrow[d, hook, "\text{open}"'] & T \arrow[d] \\ X \arrow[r, "f"] & S \end{tikzcd}$$

<u>Rem.</u> Every sq-zero thickening has a pd-str. given by $\gamma_n = 0$ for $n \geq 2$.

<u>Rem.</u> On $(\mathbb{Z}/4\mathbb{Z}, (2))$, this is <u>not</u> standard pd-str. $\gamma_4(2) = 3$.

③ (Assuming f is formally smooth)

An association $\forall (U \hookrightarrow T) \in \mathrm{Crys}^2(X/S)$ a q-coh. $\mathcal{F}_{U \hookrightarrow T} / T$ s.t.

$\mathcal{F}_{X \overset{id}{\to} X} \simeq \mathcal{F}$, $g^* \mathcal{F}_{U \hookrightarrow T} \simeq \mathcal{F}_{U' \to T'}$,
$$\begin{tikzcd} U' \arrow[r] \arrow[d] & T' \arrow[d, "g"] \\ U \arrow[r] & T \end{tikzcd}$$
+ cocycle

localize
② $\Leftrightarrow$ ③ : <u>Claim.</u> $(X \xrightarrow{id_X} X) \in \mathrm{Crys}^2(X/S)$ is weakly final.

$\Uparrow$

formal smoothness of $f : X \to S$

Given $\alpha_1, \alpha_2 \rightsquigarrow$

$$\begin{array}{ccc} \mathcal{U} & \longrightarrow & T \\ \downarrow & & \downarrow (\alpha_1, \alpha_2) \\ X & \longrightarrow & X^{(2)} \end{array}$$

Given ②, 1. lift $\alpha : T \hookrightarrow X$, define $\mathcal{F}_{\mathcal{U} \to T} := \alpha^* \mathcal{F}$.

2. Given two diff. lifts $\alpha_1^* \mathcal{F} \simeq \alpha_2^* \mathcal{F}$ by pulling back $p_1^* \mathcal{F} \simeq p_2^* \mathcal{F}$ along $(\alpha_1, \alpha_2) : T \to X^{(2)}$

3. Cocycle condition for $\alpha_1, \alpha_2, \alpha_3$ follows from cocycle cond. on $X^{(3)}$.

③ $\Rightarrow$ ② evaluate on

$$\begin{array}{ccc} \mathcal{U} & \hookrightarrow & T \\ \| & & \\ X & & X^{(2)} \end{array}$$

① $\Rightarrow$ ③

$$\begin{array}{l} X = \operatorname{Spec} B \\ \downarrow \text{ smooth} \\ S = \operatorname{Spec} A \end{array}$$

local coord

$x^1, \cdots, x^d$

$dx^1, \cdots, dx^d$

$\updownarrow$

$\partial_1, \cdots, \partial_d$

Given $B \overset{f}{\underset{g}{\rightrightarrows}} C$, $I \subset C$, $I^2 = 0$, $f \equiv g \pmod{I}$

$\mathcal{F} \longleftrightarrow M$ a B-module

$\nabla : M \longrightarrow M \otimes_B \Omega_{B/A}$, $\nabla(m) = \sum_{i=1}^{d} \nabla_i(m) \otimes dx^i$

$\nabla_i(bm) = (\partial_i b) \cdot m + b \nabla_i(m)$

What is $M \otimes_{B,f} C \xrightarrow{\sim} M \otimes_{B,g} C$?

A. $m \otimes 1 \longmapsto m \otimes 1 + \sum_{i=1}^{d} \nabla_i(m) \otimes h(x^i)$, $\quad h(b) = f(b) - g(b) \in I$.

Why well-defined?

$$\begin{aligned} bm \otimes 1 \longmapsto\ & bm \otimes 1 + \sum \nabla_i(bm) \otimes h(x^i) = m \otimes g(b) + \sum \nabla_i(m) \otimes g(b) h(x^i) \\ & \qquad + \sum m \otimes g(\partial_i b) h(x^i) \end{aligned}$$

$$bm \otimes 1 = m \otimes f(b) \longmapsto m \otimes f(b) + \sum \nabla_i(m) \otimes h(x^i) f(b)$$

(the two results are equal: $//$)

Key point. $f(b) = g(b) + \sum g(\partial_i b) h(x^i)$

Relax cond. that $I^2 = 0$. instead I is nilpotent

real Taylor expansion.

$$f(b) = \sum_{k_1, \dots, k_d = 0}^{\infty} g\left(\frac{\partial_1^{k_1} \cdots \partial_d^{k_d} b}{k_1! \cdots k_d!} \right) h(x^1)^{k_1} \cdots h(x^d)^{k_d}$$

For $M \otimes_{B,f} C \xrightarrow{\sim} M \otimes_{B,g} C$

$$m \otimes 1 \longmapsto \sum_{k_1, \dots, k_d = 0}^{\infty} \nabla_1^{k_1} \cdots \nabla_d^{k_d}(m) \otimes \overbrace{\frac{h(x^1)^{k_1}}{k_1!} \cdots \frac{h(x^d)^{k_d}}{k_d!}}^{h(x^1)^{[k_1]} \cdots h(x^d)^{[k_d]}}$$

This can be defined if

(I) $\nabla_i \nabla_j = \nabla_j \nabla_i \iff \nabla$ is integrable

(II) pd-structure γ on (C, I)

either

(II-1) (C, I) is pd-nilpotent — Def PD-nilp means $\exists\, C$ s.t. $x_1^{[k_1]} \cdots x_m^{[k_m]} = 0$ for $k_1 + \dots + k_m > C$

(II-2) ∇ is nilpotent — Def ∇ is nilp when $\exists\, N$ s.t. $\forall\, m \in M$, $\nabla_1^{k_1} \cdots \nabla_d^{k_d}(m) = 0$ if $k_1 + \dots + k_d > N$

Rem. $(\mathbb{Z}/2^n, (2))$ not PD-nilp.

Conclusion. If $X \to S$ smooth, $(\mathcal{E}, \nabla)$ nilp. integrable conn.

$\rightsquigarrow$ a crystal on $\mathrm{Crys}(X/S)$

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a sheaf of $\mathcal{O}_{X/S}$-mod on $\mathrm{Crys}(X/S)$

s.t. ① $(U \hookrightarrow T) \rightsquigarrow \mathcal{F}_{U \to T}$ a q-coh.

② $\begin{array}{c}(U' \hookrightarrow T') \\ \downarrow \quad \downarrow g \\ (U \hookrightarrow T)\end{array}$ $\qquad g^* \mathcal{F}_{U \to T} \xrightarrow{\simeq} \mathcal{F}_{U' \to T'}$

$\sim$

Functoriality (All schemes are locally torsion)

(S, I, γ) base pd-scheme

$X \to S$ s.t. "γ extends to X"

i.e. $\exists$ pd str. on $I\mathcal{O}_X$ compatible with γ

$\mathrm{Crys}(X/S)$: objects are
$$\begin{tikzcd} U \arrow[r, hook, "pd"] \arrow[d, "open"'] & T, \delta \arrow[d] \\ X \arrow[r] & S \end{tikzcd}$$
, $V(J) = U$

s.t. "δ is compatible with γ"

$I\mathcal{O}_T + J$ has a pd-str.

Consider
$$\begin{tikzcd} X' \arrow[r, "g"] \arrow[d] & X \arrow[d] \\ S' \arrow[r, "f"] & S \end{tikzcd}$$
where f is a pd-morphism. $f^* I \subset I'$, γ, γ' compatible

in Crys(X'/S')

(*)
$$\begin{tikzcd}
U' \arrow[rr, hook] \arrow[dr] \arrow[dd] & & T', \delta' \arrow[dr] \arrow[dd] & \\
& U \arrow[rr, hook] \arrow[dd] & & T, \delta \arrow[dd] \\
X' \arrow[rr] \arrow[dr] & & S' \arrow[dr] & \\
& X \arrow[rr] & & S
\end{tikzcd}$$

in Crys(X/S)

$$(X'/S')^{pre}_{crys} \underset{g^{pre*}_{crys}}{\overset{g^{pre}_{crys*}}{\rightleftarrows}} (X/S)^{pre}_{crys}$$

Given $F \in (X'/S')^{pre}_{crys}$

$$\left(g^{pre}_{crys*} F\right)(U \to T, \delta) = \varprojlim_{\substack{(*) \text{ with} \\ \text{fixed } (U,T,\delta)}} F(U' \hookrightarrow T', \delta')$$

Given $\mathcal{G} \in (X/S)^{pre}_{crys}$

$$\left(g^{pre*}_{crys} \mathcal{G}\right)(U' \hookrightarrow T', \delta') = \varinjlim_{\substack{(*) \text{ with} \\ \text{fixed } (U',T',\delta')}} \mathcal{G}(U, T, \delta)$$

$$g^{pre*}_{crys} \dashv g^{pre}_{crys*}$$

Claim. g^{pre}_{crys*} actually sends sheaves to sheaves.

$T = \bigcup T_i$ and $\forall\, T' \to T \rightsquigarrow \underline{T' = \bigcup T_i'}$

$g^{pre}_{crys*} F \longleftarrow$ F a sheaf has gluing property here

has gluing property $\square$

Def $g_{crys\,*} = g^{pre}_{crys\,*}$

$g^{*}_{crys} = \left(g^{pre}_{crys\,*}\right)^{\#}$ } adjoint

Claim. $g^{pre\,*}_{crys}$ is exact $\Rightarrow$ g^{*}_{crys} is exact

$\Uparrow$

Claim. If we fix U', T', δ', the cat. of ⊛ is cofiltered ($\Rightarrow$ taking filtered colimit)

Say

$$\begin{tikzcd} U_1 \arrow[r] \arrow[d] & T_1 \arrow[dr] & U_2 \arrow[r, hook] \arrow[dll] & T_2 \arrow[dl] \\ X \arrow[rr] & & S & \end{tikzcd}$$

Want.

$$\begin{tikzcd} & U_3 \arrow[r] \arrow[dl] \arrow[dr] & T_3 \arrow[dll] \arrow[dr] & \\ U_1 \arrow[r] & T_1 & U_2 \arrow[r] & T_2 \end{tikzcd}$$

Consider. $U_3 = U_1 \cap U_2$, $U_3 \overset{\Delta}{\hookrightarrow} T_1 \times_S T_2$ locally closed

$$\begin{array}{cc} T_1 & T_2 \\ \uparrow & \uparrow \\ pd & pd \end{array}$$

Claim. $\exists$ a universal $U_3 \hookrightarrow T_3 \to T_1 \times_S T_2$ such that

$+$ $U_3 \to T_3$ pd-str.

① $T_3 \to T_1$ is a pd morphism

② $T_3 \to T_2$ —"—

③ $U_3 \hookrightarrow T_3$ compatible with pd-str. on S.

Univ. prop $\Rightarrow$

$$\begin{tikzcd} U' \arrow[r] \arrow[d] & T' \arrow[d] \\ U_3 \arrow[r] & T_3 \end{tikzcd}$$

Affine locally

$$\begin{tikzcd} A \otimes B \arrow[r, two heads] \arrow[d] & C = (A \otimes B)/I \\ D \arrow[ur, "\text{nilp. quotient}"'] & \end{tikzcd}$$

pd-thickening

$D = \dfrac{\text{free alg. gen. by } \gamma_\bullet(\text{elts in } I)}{\text{usual pd-rel.}}$

$x \in I \subset A$

$\gamma_n(x \otimes 1) = \gamma_n(x) \otimes 1$

$\vdots$